

Q1

Facebook has ^{3.07}~~2.7~~ billion users. If a programmer wants to use an unsigned int as user-id, what's the smallest sized int should they use?

☐ 1-byte

☐ 2-byte

☐ 3-byte $\rightarrow 24b \rightarrow 0 \sim 2^{24} - 1$

☒ 4-byte $\rightarrow 32b \rightarrow 0 \sim 2^{32} - 1$

☐ 8-byte

$$2^{24} - 1 < 3.07 \times 10^9 < 2^{32} - 1$$

$$(2^{10} = 1024 \approx 10^3)$$

Q2

Which of the following signed 1-byte int (in binary format) is the smallest?

☐ 00000000

☒ 10000001

☐ 11111111

☐ 00000001

☐ 10000011

☐ 01111110

• For signed int, convert it into a decimal number:

• For i-th bit, (i-th bit) * (+/- 2^i).

• highest bit: *(- 2^i), others: *(+ 2^i)

• Smallest:

• Highest bit is 1

• For other bits(positive), pick the smaller one

Q3

Convert bit pattern 10011110 to hex notation. To facilitate autograding, you must prefix your answer with 0x and use lower case.

1001 1110

0x9e

9 e

Q4

Which of the following statements are true?

☐ Programmers should always use 8-byte for their integers.

☒ A 64-bit processor supports computation on signed/unsigned 8-byte integers.

☒ Byte is the smallest unit for storing and transferring information in a computer.

☐ Bit is the smallest unit for storing and transferring information in a computer.

Q5

Which of the following 1-byte **signed** addition operation will overflow?

☐ 0xff + 0xfe $0xf+0xf=0x\cancel{e}$

☐ 0x1f + 0xff $pos + neg$

☒ 0x71 + 0x70

☐ 0x05 + 0xfe $pos + neg$

☒ 0x80 + 0x8f $8+8=0x\cancel{10}$

pos + neg never overflow
 0 ~ 7 0b0... positive
 8 ~ f 0b1... negative

Q6

Signed one-byte integers a, b have bit pattern 0xf2 and 0xfe respectively. What is the bit pattern of signed one-byte integer $c=a-b$? (Please write your answer in hex, and prefix it with 0x)

0xf4

$\downarrow$ $\downarrow$
 -14 -2
 $-14 - (-2) = -12$

Q7

Suppose the byte values stored at memory address a , $a+1$, $a+2$, $a+3$, $a+4$, $a+5$, $a+6$, $a+7$ are 0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0x07, 0x08 respectively. If a Little-Endian processor is to load a 4-byte integer from memory at address a into a 4-byte register, what's the 4-byte register value after the load? (Please write your answer in hex, and prefix it with 0x)

0x04030201

a	a+1	a+2	a+3	a+4	a+5	a+6	a+7
0x01	0x02	0x03	0x04	0x05	0x06	0x07	0x08

$\leftarrow$

Q8

Which of the following is a **normalized** exponential representation in either binary or decimal?

☐ $(0.11)_2 * 2^1$
☒ $(1.00)_2 * 2^{-10}$
☐ $(10.11)_2$
☐ $(78.5)_{10} * 2^{10}$
☒ $(7.85)_{10} * 10^1$

$$\pm x.yz \dots \times b^{\pm e}$$

$$x \neq 0$$

Q9

Which of the following statements are true about IEEE FP32 (aka single-precision)?

☐ The number zero is represented in normalized encoding

☒ The number zero is represented in denormalized encoding

☒ All denormalized numbers are closer to zero than normalized numbers

☐ Some but not all denormalized numbers are closer to zero than normalized numbers.

☒ The exponent value (E) in denormalized encoding is $1-127 = -126$.

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normalized: $1 \leq e \leq 254$

denormalized: $e = 0$

$$0.abc \times 2^{-126} < 1.0 \times 2^{-126}$$

Q10

What's the value of IEEE FP32 (aka single precision) with bit pattern

0xc0600000?

Note that FP32 has 8 bit for exponents (bias=127) and 23 bit for mantissa. (Give your answer in the form of regular decimal fractional notation xxx.yyy with no leading nor trailing zeros)

-3.5

$$\begin{array}{c|c|c|c} 1 & 100 & 0000 & 0110 & 0000 & \dots \\ \hline s & & e & & m & \end{array}$$

$$(128-127)=1$$

1.110...

$$\downarrow$$

$$1.75$$

$$-1.75 \times 2^1 = -3.5$$